A265 – Redshift Reinterpreted: Static, Achromatic, and Without Expansion

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Contents

.1 Purpose

[STIPULATION] This document develops three related theses:

(1) Redshift as a static geometry effect. In FFGFT z arises not from space expansion but from energy loss of the photon in the ξ-field along the light path: $1 + z = e^{\xi x}$.

(2) Achromaticity. The mechanism is purely geometric – it stretches all wavelengths equally. Earlier presentations of chromatic T0 redshift in the corpus are recorded as a correction in the correction programme.

(3) The SI anchor of the sector. Both FFGFT and the Standard Model need an empirical anchor for the transition from natural units to measurable SI quantities in cosmology. In FFGFT this anchor is the ratio $m_e c^2/k_B T_{\mathrm{CMB}}$ – a measured number, not pure geometry. The SM has the analogue: H_0 in km/s/Mpc presupposes c, k_B, and the megaparsec definition, which are themselves historical SI conventions.

.2 The Static Redshift

[B] A photon loses energy linearly with distance travelled in the static ξ-field:

$$\frac{dE}{dx} = -\xi E \;\Rightarrow\; E(x) = E_0\, e^{-\xi x}.$$

The redshift:

$$\boxed{1 + z = e^{\xi x}, \qquad z \approx \xi x \;\text{ (small } z\text{)}}$$

No scale factor $a(t)$, no Big Bang, no expanding space. Light travel time to an object with redshift z:

$$\tau(z) = \frac{\ln(1+z)}{H_0^{\mathrm{T0}}}.$$

The function grows without bound – there is no Big Bang ceiling.

[K] Comparison with the Hubble law. For small z: $z \approx \xi x$. Hubble law: $z = H_0 d/c$. Identification: $H_0 = \xi c$. The T0 formulation uses the Hubble energy $E_H = E_0 \cdot \xi^{41/4}$, from which $H_0^{\mathrm{T0}} = E_H/\hbar \approx 66.2$ km/s/Mpc ($-1.9\,\%$ vs. Planck). The exponent $41/4$ is not derived from ξ alone – that is the named open point.

.3 Achromaticity: Correction of the Earlier Presentation

[B] The mature FFGFT mechanism is *fractal path elongation*: the path through the recursive time geometry is stretched by a factor that depends solely on the distance – not on the wavelength of the photon. Hence $1 + z = e^{\xi x}$ holds for *all* wavelengths with the same factor.

[K] Physical reason. A *geometric* stretch factor acts on wavelength and pulse duration in the same ratio – just as gravitational redshift and standard cosmological redshift are achromatic. A wavelength-dependent term would arise only from a dispersive medium effect, not from pure geometry.

[K] Consequence. The observed cosmological redshift is achromatic to high precision – FFGFT gets this right. The earlier presentation of chromatic T0 redshift in earlier versions of the legacy corpus is superseded; the correction is recorded as a correction entry.

.4 The SI Anchor: Where the Circularity Sits

[B] This is the methodologically most important section. FFGFT derives H_0 from ξ and E_0. The derivation ends with:

$$H_0^{\mathrm{T0}} = \frac{E_H}{\hbar}, \qquad E_H = E_0 \cdot \xi^{41/4}.$$

Apparently a pure ξ-derivation. But $E_0 = \sqrt{m_e m_\mu} = 7.398$ MeV is a *measured* quantity (A130), and the transition to km/s/Mpc needs $\hbar$ and the megaparsec definition – both SI conventions.

[K] The remaining anchor. After cancelling the $\hbar c$ factors (which appear in both length scales L_ξ and $\bar{\lambda}_e$) what remains is:

$$\boxed{\frac{m_e c^2}{k_B T_{\mathrm{CMB}}} = 2.176 \times 10^9}$$

Here c^2 (mass → energy) and k_B (kelvin → energy) sit – that is a ratio of two *measured* quantities: electron rest energy to CMB temperature. The bridge formula is thus geometry form ($K_{\mathrm{geo}} \cdot \xi^{41/4}$) **times** SI anchor ($m_e c^2/k_B T_{\mathrm{CMB}}$) – not a pure geometry identity.

[B] Where the exponent changes. The same ratio sits in the exponent itself: referenced to the Planck scale instead of E_0, the exponent becomes $63/4 = 41/4 + 11/2$, where $11/2$ encodes the SI scale conversion $\ell_P \leftrightarrow E_0$. The factor never disappears – it only changes location.

Reference scale	Exponent	SI factor explicit as
E_0 (electron/CMB)	$41/4$	Ratio $m_e c^2/k_B T_{\mathrm{CMB}}$
ℓ_P (Planck)	$63/4 = 41/4 + 11/2$	Additional exponent $11/2$

.5 The SM Has the Same Problem

[B] The Standard Model gives H_0 in km/s/Mpc. This unit presupposes: c (speed of light in km/s), the megaparsec definition ($1\,\mathrm{Mpc} = 3.0857 \times 10^{22}$ m, historically from parallaxes), k_B for temperature measurements, and the SI 2019 reform that fixes c, $\hbar$, k_B, and e by definition. These quantities are not independent of one another – the measurement chain is self-referential.

[B] Concretely: $T_{\mathrm{CMB}} = 2.7255\,\mathrm{K}$ (Fixsen 2009) is linked to energy via k_B; $H_0 \approx 67.4$ km/s/Mpc is linked to a length via c and Mpc; the ratio $m_e c^2/k_B T_{\mathrm{CMB}}$ connects particle physics with cosmology – and this ratio has no place in the SM basic theory. The SM calls it a measurement; FFGFT attempts to understand it from ξ, and comes closer, but not all the way (the exponent $41/4$ is missing).

[S] The honest comparison. Both frameworks – ΛCDM and FFGFT – need an empirical anchor for cosmology that lies outside the theoretical core. In the SM that is H_0 itself (a measurement, not a derivation). In FFGFT it is $m_e c^2/k_B T_{\mathrm{CMB}}$ plus the not-yet-derived exponent $41/4$. FFGFT is one step further here – but not yet completely parameter-free.

.6 Cosmological Degeneracy

[K] FFGFT and ΛCDM are *observationally degenerate* on the standard tests: both reproduce the same fluxes, angular sizes, and time dilations. The key points:

Supernova time dilation: DES measures $b = 1.003 \pm 0.011$ (White et al. 2024, Dark Energy Survey, MNRAS). The fractal path elongation stretches wavelength *and* pulse duration in the same ratio $\Rightarrow b = 1$. The test only excludes non-dilating tired light ($b = 0$) – FFGFT predicts $b = 1$ and is not affected.

Hubble tension: In FFGFT H_0 is not an expansion rate but the geometric energy-loss coefficient. Different measurement methods probe different scales; systematic variations are geometrically expected. The tension 67.4 (Planck 2018) vs. 73.0 km/s/Mpc (SH0ES) is not a fundamental problem in FFGFT.

[S] Honest situation. The degeneracy cuts both ways: the data *fit* FFGFT (no wrong prediction), but they do not *force* it. The case for FFGFT rests on parsimony (no Λ, no dark energy) and on the ξ-derivation of the constants – not on a single discriminator.

.7 Verification Script

- Numerical verification: $1 + z = e^{\xi x}$, achromaticity (same stretch factor for $\lambda_1 \neq \lambda_2$), SI anchor $m_e c^2/k_B T_{\mathrm{CMB}}$, exponent table $41/4$ vs. $63/4$: `python/A_Serie_Skripte/a265_rotverschiebung.py`

.8 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
z	Cosmological redshift; $1 + z = \lambda_{\text{obs}}/\lambda_{\text{emit}}$
$E_H = E_0 \cdot \xi^{41/4}$	Hubble energy in FFGFT
$H_0^{\text{T0}} \approx 66.2$ km/s/Mpc	FFGFT value of the Hubble constant
$\tau(z) = \ln(1 + z)/H_0^{\text{T0}}$	Light travel time in FFGFT
$m_e c^2/k_B T_{\text{CMB}}$	Empirical SI anchor of the cosmological sector

.9 Open Questions

The exponent $41/4$ is not derived from ξ alone. It is calibrated to the measured H_0. As long as it is missing, FFGFT in the cosmological sector is not parameter-free.

The anchor $m_e c^2/k_B T_{\text{CMB}}$ needs a geometric explanation. Why precisely the ratio of electron rest energy to CMB temperature appears as the natural transition is an open structural question.

BAO and CMB peaks have not yet been derived from FFGFT. The degeneracy claim presupposes that the same angular scales are reproducible; the proof in FFGFT's own formalism is outstanding.

.10 Supersedes

This document subsumes: Dok. 026 (static redshift as geometric path elongation, FEM simulation, Hubble energy $E_H = E_0 \xi^{41/4}$, derivation $H_0 \approx 66.2$ km/s/Mpc), Dok. 280 (static redshift, achromaticity as correction K6, z →time as ΛCDM transfer, cosmological degeneracy, no Big Bang), Dok. 279 (Casimir–H_0 bridge formula, SI anchor $m_e c^2/k_B T_{\text{CMB}}$, exponent $41/4$ vs. $63/4$, location of SI factor).

.11 Use of AI Tools

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